

Quantitative Investing

PROBLEM SET 5 — PART I: BUSINESS CYCLE VARIATION

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DATA AND CONVENTIONS

- **Sample.** Monthly data from 1926-07 to 2026-06 (UMD and the momentum portfolios start 1927-01). The NBER dummy marks 201 recession months across 16 recessions.
- **Returns.** All figures are percent per month. Portfolio returns are in excess of the T-bill; RMRF is already excess, and SMB, HML and UMD are zero-cost long-short returns, so they are used as given.
- **Inference.** Residuals cluster in recession episodes, so t -statistics use Newey–West standard errors with 12 lags.
- **Shared code.** All parts load data through `common/data.py` and run Fama–MacBeth through `common/fmb.py`, so lags, samples and beta definitions are identical across Questions (a)–(h).

QUESTION (A)

We estimate $R_t = a + b D_t + \varepsilon_t$, where $D_t = 1$ in NBER recession months, so a is the non-recession mean and $a + b$ the recession mean.

Table 1: Factor returns in and out of recessions (% per month)

	Non-recession a	Recession $a + b$	Difference b	$t(b)$
RMRF	0.93	−0.47	−1.40	−2.50
SMB	0.19	0.03	−0.16	−0.82
HML	0.40	0.15	−0.25	−0.75
UMD	0.71	0.22	−0.49	−1.00

Newey–West (12 lags) t -statistics. UMD starts 1927-01.

- **What does this tell you?** Only the market is reliably worse in recessions; SMB, HML and UMD give up some return but none of the drops is distinguishable from zero. All three factors still earn a positive average return in recession months, while the market loses money. So the factor premia are not mainly a reward for losing money in recessions, at least not in the way the market premium is.
- **Based on these results, do you believe SMB or HML is more consistent with a risk-based explanation?** SMB is, but only weakly. Its premium all but disappears in recessions, so small stocks fail to pay off exactly when times are bad, which is what a risk story needs. HML instead keeps a positive return in recessions, in line with Lakonishok, Shleifer and Vishny’s (1994) finding that value does not underperform in bad states. With 201 recession months neither difference is significant, so we read this as a lean rather than a verdict.

QUESTION (B)

Table 2: Corner portfolio excess returns in and out of recessions (% per month)

	Non-recession a	Recession $a + b$	Difference b	$t(b)$
<i>Size-BE/ME corners</i>				
Small Growth	0.83	-0.63	-1.46	-1.30
Small Value	1.69	-0.38	-2.07	-2.62
Big Growth	0.90	-0.34	-1.25	-2.21
Big Value	1.24	-0.12	-1.35	-1.51
<i>Size-momentum corners</i>				
Small Losers	0.80	-0.26	-1.06	-1.17
Small Winners	1.84	-0.22	-2.06	-2.98
Big Losers	0.58	-0.83	-1.42	-1.66
Big Winners	1.24	-0.40	-1.64	-2.73
<i>Within-size spreads</i>				
Small Value – Growth	0.87	0.25	-0.62	-1.03
Big Value – Growth	0.34	0.23	-0.11	-0.14
Small Winners – Losers	1.04	0.04	-1.00	-1.29
Big Winners – Losers	0.64	0.39	-0.25	-0.40

Excess returns over the T-bill. Newey–West (12 lags) t -statistics. Momentum portfolios start 1927-01.

- **Do you find stronger cyclical patterns for these extreme portfolios than for the factors?** Yes, but mostly because the corners are long-only portfolios that carry the market's recession losses. Every corner falls by one to two percent per month in recessions, and the high-return corners (Small Value and both Winner portfolios) fall the most. Once we net out the market by taking spreads within a size group, the pattern shrinks back to factor-sized, insignificant drops that stay positive in recessions. The extreme portfolios are more cyclical, but the value and momentum premia themselves are not.